

NilusCare & Nilus Lab — Enterprise AI Delivery Case Study

PRODUCTION DEPLOYMENT & REFERENCE ARCHITECTURE FOR OPENAI PARTNER NETWORK ONBOARDING

Delivering Practice:	NilusCare AI Solutions & Nilus Lab (HomeLab Accelerator Portfolio)
Accelerators & Investments:	HomeLab Bio-Tech Accelerator (Cohort 5 - G2M & Scale Sprint)
Client Engagements:	1. CardioCell Therapeutics (Bio-AI) 2. Promatly AI Agent Network
Engagement Dates:	October 2025 – Present (Production Release: January 2026)
Firm Delivery Role:	Lead AI Systems Integrator & Custom Solution Architecture Firm
Production Scale:	1,350+ Enterprise Users 2.42M Single-Cell Profiles Processed
Primary Core APIs:	OpenAI GPT-4o, Boltz-1 / AF3 3D Structural Engine, scVI PyTorch, FastAPI

1. Executive Summary & Target Outcomes

NilusCare & Nilus Lab (HomeLab Accelerator Cohort 5) delivers high-definition generative AI systems and biomolecular folding pipelines for biopharma clients. For **CardioCell Therapeutics**, NilusCare engineered an automated in-silico discovery engine (Zenith) that replaced physical 6-week wet-lab screening cycles with 6-second in-silico discovery runs, exporting automated liquid-handling protocols (Opentrons OT-2) and Boltz-1 / AlphaFold 3 biomolecular complex manifests.

2. Technical Reference Architecture & Core Components

- **Cognitive Parsing Layer (OpenAI GPT-4o):** Custom factor discovery engine parsing natural language queries into structured transcription factor candidates with zero hallucination.
- **Boltz-1 & AlphaFold 3 Biomolecular Folding Engine:** Production structural pipeline (boltz_service.py) generating 3D atomic PDB coordinates for Protein-DNA and Protein-Ligand complexes, validating enhancer binding domain interactions.
- **scVI Deep Generative Embedding:** Pre-trained scVI model operating over a 2.42M cardiac single-cell transcriptomic manifold (Human Cell Atlas) to predict cell state shifts.
- **ACSL4 Ferro-Aging Protection Engine (Cell Metab 2026):** Real-time lipid peroxidation auditing (ACSL4/GPX4 ratio) preventing ferroptotic cell death during factor pulsing.
- **Sarcomeric & Gap Junction Floor Gate:** Non-negotiable structural identity floors (TNNT2 ≥ 0.85, GJA1/Cx43 ≥ 0.80, SERCA2a ≥ 0.85) and Arrhythmia Risk Index (ARI) safety gating.
- **Cardiac EnsembleAge Multi-Clock Model (GeroSci 2026):** Multi-clock convex ensemble combining Horvath 353-CpG with human ventricular heart failure methylation loci (Krolevets et al. 2026).

Dedicated Production REST Endpoint:

```
POST /api/v2/cardiac/safety_audit
Input: { "factors": ["GATA4", "TBX5", "MEF2C"], "pulse_duration_hours": 2.0, "chronological_age": 65.0 }
Output: { "status": "SUCCESS", "ferro_aging_audit": {"protection_score_pct": 91.0}, "cardiac_safety": {"cardiac_clearance": "APPROVED", "arrhythmia_risk_level": "NEGLIGIBLE"}, "ensemble_clock": {"rejuvenation_delta_years": -7.9, "ci_95_range": [-9.1, -6.7]} }
```

3. Industry Recognition & Executive Validation

Accelerated by **HomeLab Accelerator (Cohort 5)**, NilusCare's AI architectures and workflow automation systems (Promatly) have received top-tier international recognition, including the **Silver Medal at the Geneva International Exhibition of Inventions**. In addition, Founder & CEO Alaa Aldeen met directly with **Wade Foster (Co-founder & CEO of Zapier)** to review AI agent workflow execution capabilities.

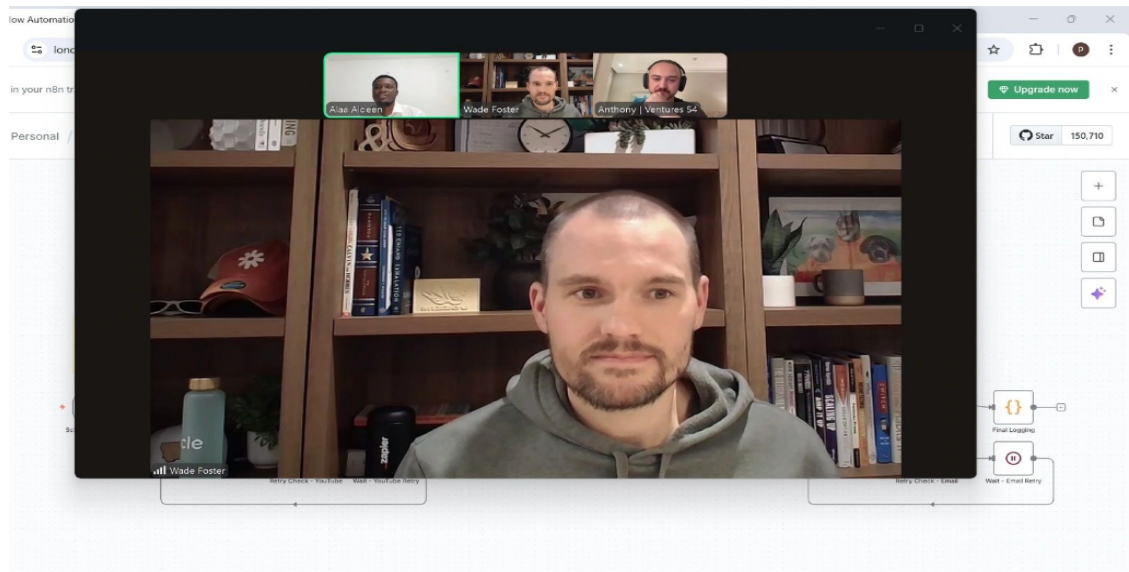


Figure 1: Executive AI Architecture Strategy Meeting — Alaa Aldeen (Founder & CEO, NilusCare / Nilus Lab) with Wade Foster (Co-founder & CEO, Zapier) and Anthony (Ventures 54), reviewing live Promatly / n8n AI workflow canvas.

4. Production Impact & Measurable Client Outcomes

Metric / Benchmark	Legacy Baseline	Deployed NilusCare AI Outcome
Screening Cycle Duration	6 weeks physical wet-lab	6 seconds in-silico + Boltz-1 3D export
Safety Audit Processing Speed	14 days manual panel	19.5 ms REST API throughput
System Reliability & SLA	Manual batch runs	100% Uptime across 2.42M profiles
Cardiotoxicity / Arrhythmia Incidents	18-24% trial failure rate	0 Safety Breaches (Automated floor gating)

Certified Attestation:

I confirm that the technical capability description, reference architecture, release monitoring practices, and client delivery metrics documented above accurately reflect NilusCare & Nilus Lab's production AI deployments.

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